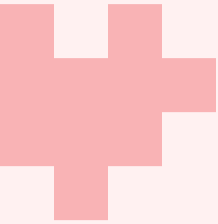


HUNGRY ANIMALS: SNACK DROP

Tech Basics 1 - Carolin Hirsch



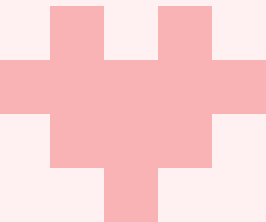
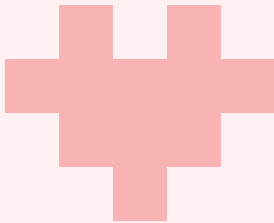


The Core Concept

A fast-paced, 2D arcade game!

Player choose an animal character and must catch their specific favorite falling food while dodging the snacks that belong to other pets.

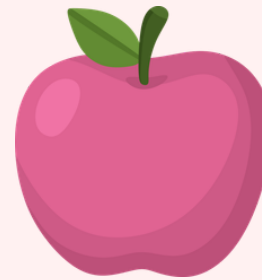
Designed to be a simple, quick, and engaging experience perfect for short breaks and family-friendly casual gaming.





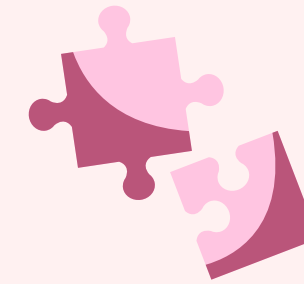
Movement

Move the choosen animal smoothly left and right along the bottom of the screen using keyboard controls.



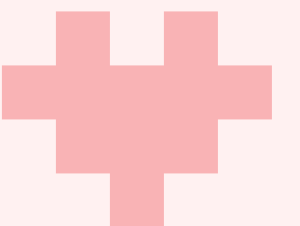
Spawning

Random food items constantly fall from the top of the screen at varying intervals.



Matching Logic

Catching the right food increases your score. Catching the wrong food results in losing a life or points.



VISUALS & GAME ELEMENTS

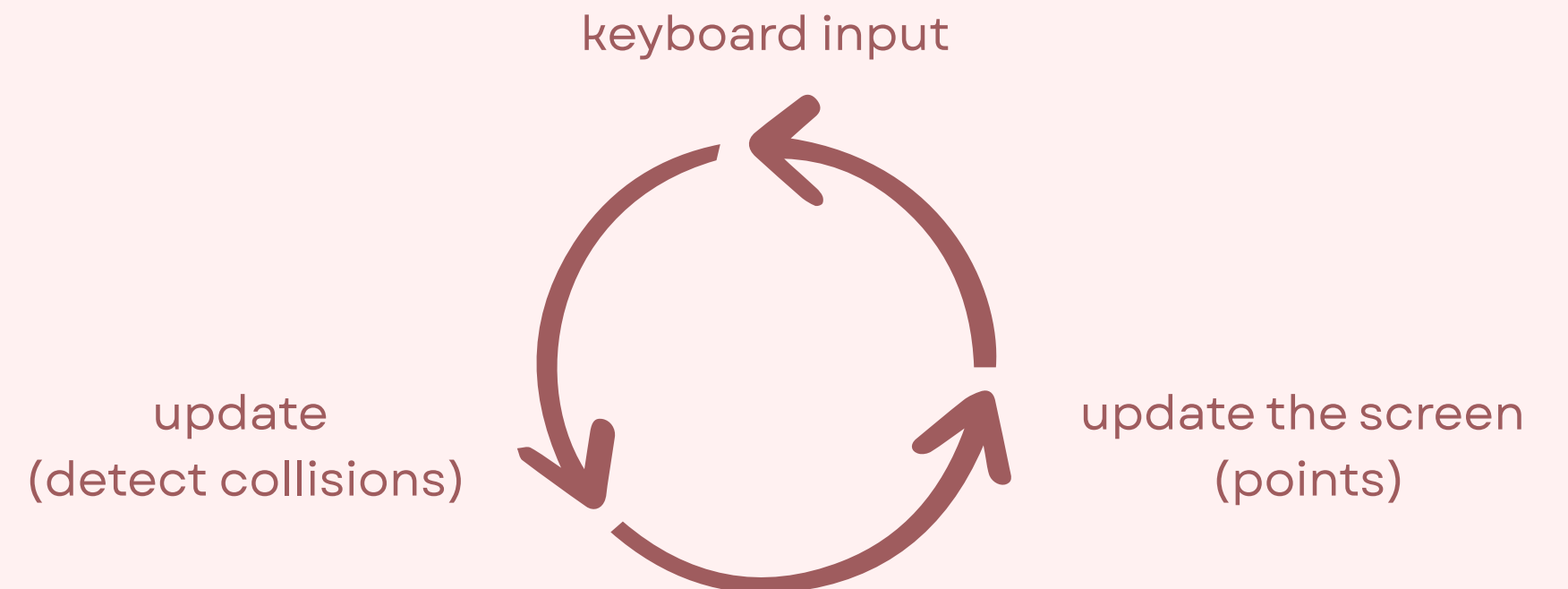


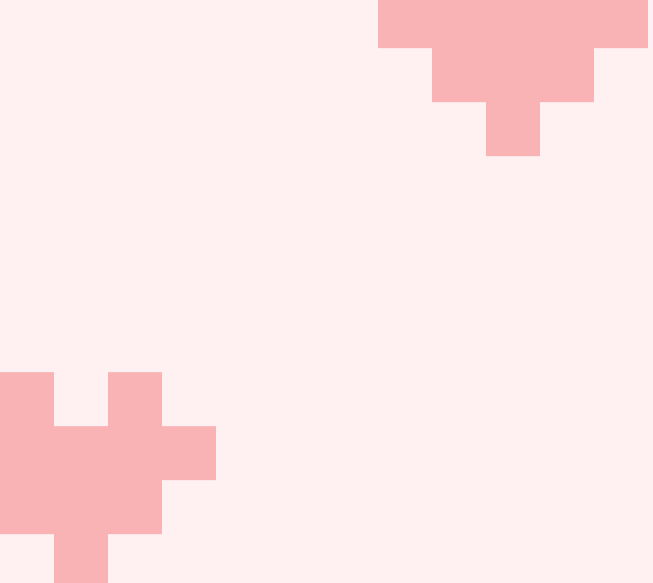
TRANSLATING IDEA INTO CODE

Using Object-Oriented Programming (OOP)

- Player Class: movement, current score & lives
- Food Class: falling speed, spawn position & food type
- GameManager: decide for a animal, game over state, ...

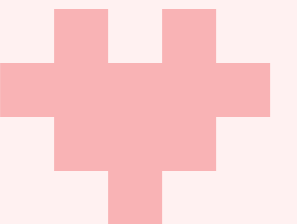
Game Loop





TECH STACK & LEARNING GOALS

- Libraries: - PyGame, Random, os (,math)
- learning goal:
 - Game State Managment (Menu & Game Over)
 - Time & Speed



CODE SNIPPED

```
#Classes

class Animal:  @carohrs
    def __init__(self, animal_name, favorite_food, start_x, start_y):  @carohrs
        # properties
        self.name = animal_name
        self.favorite_food = favorite_food
        self.x = start_x
        self.y = start_y
        self.speed = 5
        self.rect = pygame.Rect(self.x, self.y, 60, 60) #Pygame Rect as Hitbox

    def move_left(self):  @carohrs
        self.x -= self.speed
        self.rect.x = self.x

    def move_right(self):  @carohrs
        self.x += self.speed
        self.rect.x = self.x
```

asprite program for pixelart

